



15 Discovery Way • Acton, MA 01720 • tel: 978.263.3584 • fax: 978.263.5086 • email: mail@acton-research.com

XENON ARC LAMP HOUSING MODEL XS-433



TABLE OF CONTENTS

I. INTRODUCTION	3
II. SAFETY CONSIDERATIONS	4
II.1. RADIATION	4
II.2. LAMP EXPLOSION	5
II.3. OZONE	5
II.4. ELECTRICAL SHOCK	6
II.5. EMI	6
II.6. HEAT	7
III. GENERAL DESCRIPTION	8
III.1. LAMP AND REFLECTOR ADJUSTMENTS	8
III.2. BUILT-IN IGNITOR	8
III.3. LAMP COOLING	8
III.4. SAFETY AND MONITORING FEATURES	8
III.5. ELAPSED TIME INDICATOR (ETI)	9
III.6. HOUSINGS WITHOUT CONDENSING LENSES	9
III.7. MOUNTING	9
IV. LAMP INSTALLATION & INITIAL OPERATION	12
IV.1. INSTALLATION OF THE LAMP	12
IV.2. ADJUSTMENT OF LAMP, MIRROR AND LENS	15
IV.2.1. GENERAL	15
IV.2.2. ADJUSTMENTS PRIOR TO OPERATION	15
IV.2.3. ADJUSTMENTS DURING OPERATION	15
IV.2.4. LAMP OPERATION AND COOLING	17
IV.3. COOLING	18
V. LAMP HOUSING OPTICS	19
V.1. COLLIMATED BEAMS	20
V.2. UNIFORMITY	21
V.3. IMAGING THE ARC	22
V.4. REAL LENSES	22
V.4.1. SPHERICAL ABBERATIONS	23
V.4.2. CHROMATIC ABBERATIONS	23
V.5. POSITIONING THE CONDENSER TO GET A COLLIMATED BEAM	24
VI. TROUBLESHOOTING	25
VI.1. INTRODUCTION	25
VI.2. PROBLEMS	25
VI.3. INPUT CONNECTOR	29
DECLARATION OF CONFORMITY	30
WARRANTY AND RETURNS	31

I. INTRODUCTION

This manual covers the 75 watt (W) Xenon Arc Lamp Housing with internal ignitor and with F/1 UV grade fused silica condenser.

The lamp housing, when used in conjunction with the appropriate power supply, is compliant with the European EMC and Low Voltage Directories and, therefore, carries the CE mark.

II. SAFETY CONSIDERATIONS

SUMMARY OF HAZARDS

There are six hazards in the operation of systems employing these lamp housings. They are:

- Radiation
- Lamp Explosion
- Ozone
- Electrical Shock
- EMI
- Heat

The interlock system is designed for your safety. Do not defeat the interlocks.

II.1. RADIATION

The high intensity UV radiation of these lamps can permanently damage the cornea, lens, and retina of the eye, even causing blindness. This damage may not be immediately apparent. The deep UV is absorbed in the cornea or eye fluids; focused UV, VIS and NIR can damage the retina. Normal blink reaction to visible light may not be adequate protection, and a beam of invisible UV or NIR (produced by spectral filtering) can be most dangerous as the blink response is not induced. UV radiation can also cause painful sunburn, and with prolonged exposure, serious burns.

Recommendations

- Never look directly into the output beam of a housing when operating a lamp.
- Never look at a specular (mirror) reflection of the beam, even for short periods of time.
- Always wear UV safety eyewear or face mask, and protective clothing for exposed areas of skin.

II.2. LAMP EXPLOSION



When cold, xenon and mercury (xenon) arc lamps are under several atmospheres of pressure and are subject to explosion due to internal strains or to physical abuse. When hot, all lamps are under a pressure of close to 100 atmospheres and subject to violent explosion.

Recommendations

- Do not handle a bare arc lamp without safety goggles and adequate protection for exposed areas of skin.
- Do not apply torque to the lamp envelope during installation or removal.
- Do not touch the lamp envelope. Fingerprints and other contaminations left on the lamp cause a deterioration of the envelope during operation and may lead to lamp explosion.
- Check the lamp polarity before starting the lamp.
- Thoroughly clean the envelope with alcohol or a dilute solution of detergent and water after installation in the housing.
- Attach a cover plate over the condenser aperture of any housing that will be used without the condenser in place before operation.

II.3. OZONE



Shortwave ultraviolet light photolyses oxygen to produce ozone. This is emitted in the cooling air stream of the lamp housing. There is no simple way of predicting the ozone concentration (or its impact on you) e.g., operation in a small enclosed area may lead to high concentrations. Operation of the same system in a large, well ventilated laboratory may not be a problem. Recommended maximum exposures are typically:

0.1 ppm for 8 hours exposure
2 ppm for a 2 hour exposure

A very sensitive nose can detect 0.015 ppm. 1 ppm produces a strong and obnoxious odor. As a rule of thumb, if you can easily smell ozone, the level is too high for prolonged exposure.

Recommendations

- Use an ozone free lamp unless you need the shortwave UV.
- Operate the system in a large ventilated area.
- Use a commercial Ozone Eater™.

II.4. ELECTRICAL SHOCK



When this lamp housing is operated as intended, the interlock system and the package design eliminate risk of electrical shock.

Use of this lamp housing with a power supply not with this housing is not recommended.

Be aware that a high transitory voltage is used to ignite the lamp and, before ignition, the lamp terminals have a potential difference of up to 200 V. These voltages are dangerous and it is wise always to approach the lamp housing as if the interlock system were not in place.

Recommendations

- Keep personnel clear of all exposed terminals.
- Before changing lamps or working on the system, check the power supply voltmeter to be sure that internal capacitors are discharged.
- Make sure all connections are securely made and check the polarity before starting a lamp.
- Do not handle the lamp leads during lamp ignition.
- Attach a cover plate over the condenser aperture of any housing that will be used without the condenser in place, before operation.

II.5. EMI



This lamp housing with its associated cables and Power Supply contains EMI (Electro Magnetic Interference) under normal operating conditions.

However, ignition of an arc lamp requires high voltage pulses to break the lamp down and a high current dump (kHz discharge) to sustain the arc. The ignition creates significant electromagnetic energy. The high level of EMI integrity of the source contains most of this transient energy, but good earth grounding, cable routing practice, and EMI shielding may be necessary to protect sensitive digital circuitry from these pulses.

Recommendations

- Start the arc lamp before powering nearby computer systems.
- Keep the computer at least 6 feet away from the ignitor/power supply.
- Use a different outlet and line for the computer and ignitor/power supply.

II.6. HEAT



These arc lamps become very hot after several minutes of operation, and remain in such state for many minutes after being shut off.

The Arc Lamp Power Supply will continue to operate the lamp housing fan to cool the lamp off after the lamp/power supply has been shut off. As long as the power supply is connected to the mains, the fan will run until the lamp has cooled down.

Recommendations

- Use the power supply circuit breaker/power switch to turn the lamp off. Do not use a switched mains receptacle since this will prevent the fan assisted lamp cool down.
- Wait for the housing fan to stop, or at least 10 minutes after turning off the lamp, before opening the lamp housing door.
- Approach the lamp as if it were hot under any circumstances.

III. GENERAL DESCRIPTION

These lamp housings are lightweight but rugged. They are excellent choices for most low to medium power laboratory lamps. You can mount them to optical tables, rails, or benches. Various models with different condensers are available. With the complete line of optical accessories described in Volume II, you can filter, focus, and steer the output to fit your application.

III.1. LAMP AND REFLECTOR ADJUSTMENTS

The position of the arc (relative to the terminals that secure the lamp) varies slightly from lamp to lamp. This is because of normal arc lamp manufacturing tolerances. These lamp housings have adjustments to place the arc of a new lamp in the same position as that of the lamp being replaced. Three fine controls position the reflector. Then, independent controls position the arc horizontally and vertically. The reflector and its controls travel with the vertical and horizontal controls. This eliminates the need for iterative lamp-reflector adjustments. See Figure 1 for location of adjusters.

III.2. BUILT-IN IGNITOR

The housing described in this manual has an ignitor built into the top of the housing. Short arc lamps require a high voltage spike, up to 40 kV for Xe lamps, to ignite. Having the ignitor built into the lamp housing minimizes unwanted radiated or conducted EMI and reduces potential exposure to the high voltage ignition energy.

III.3. LAMP COOLING

The built in fan and housing baffles maintain the proper operating temperature for arc lamps, when operated in normal laboratory ambient. Overheating due to blocking of the cooling vents or an inoperative fan, activates a thermostat interlock that disables the power supply.

The fan speed is controlled to keep the lamp at a constant operating temperature. The fan starts slowly when the lamp is ignited, speeds up as the lamp approaches the operating temperature, runs at (almost) constant speed after the lamp has warmed up and continues to run after the lamp has been turned off until the lamp has cooled down.

The cooling fan receives its power directly from the power supply.

NOTE: The fan cooling will not be effective if the condenser is removed and the aperture is not covered.

III.4. SAFETY AND MONITORING FEATURES

These lamp housings incorporate safety interlocks. When used with the Power Supply provided with the housing, the lamp automatically shuts off if the housing door is opened or the housing overheats. If you use your own power supply, we strongly recommend that you utilize this low voltage interlock system for safety.

III.5. ELAPSED TIME INDICATOR (ETI)

The ETI is a six digit LCD meter located on the side of the housing just above the fan. The time accumulates, in one-tenth hour increments up to 99,999.9 hours, whenever the lamp is running and the AC line is connected to the lamp housing. An hourglass symbol at the left of the display flashes when the lamp is on. The accumulated run time is saved when power is removed from the lamp housing. The ETI can be reset to 0.0 hours by momentarily pressing the reset switch located below and to the right of the ETI. The reset switch is recessed into the housing so you will need to use a tool (like a pencil) to access it.

The Elapsed Time Indicator (ETI) is a convenient way to monitor the time that a lamp has been running. Arc lamps deteriorate with usage and should be replaced after the rated life to avoid rapid reduction in output and potential dramatic lamp failure. See Oriel Volume II or the lamp manufacturer's literature for information on lamp life. Typically, the ETI would be reset each time the lamp is changed.

III.6. HOUSINGS WITHOUT CONDENSING LENSES

If the condenser lens is removed, there will be a large aperture in the front of the housing. This opening compromises the integrity of the housing and exposes the operator to the potential hazards of electrical shock and lamp explosion and affects the lamp cooling process. The condenser aperture **must** be securely covered before operating the lamp.

III.7. MOUNTING

The lamp housings are supplied with four adjustable feet for free standing use. They allow 0.63 inch (16 mm) height adjustment. You can also mount the housing to tables, carriers, or translators with tapped holes on 1 inch or 25 mm centers. To do this you should remove the feet and use the three mounting points and screw slots. If you have a flat enough surface, you can unscrew the three mounting pads and use the baseplate directly. See Figure 2 for baseplate dimensions.

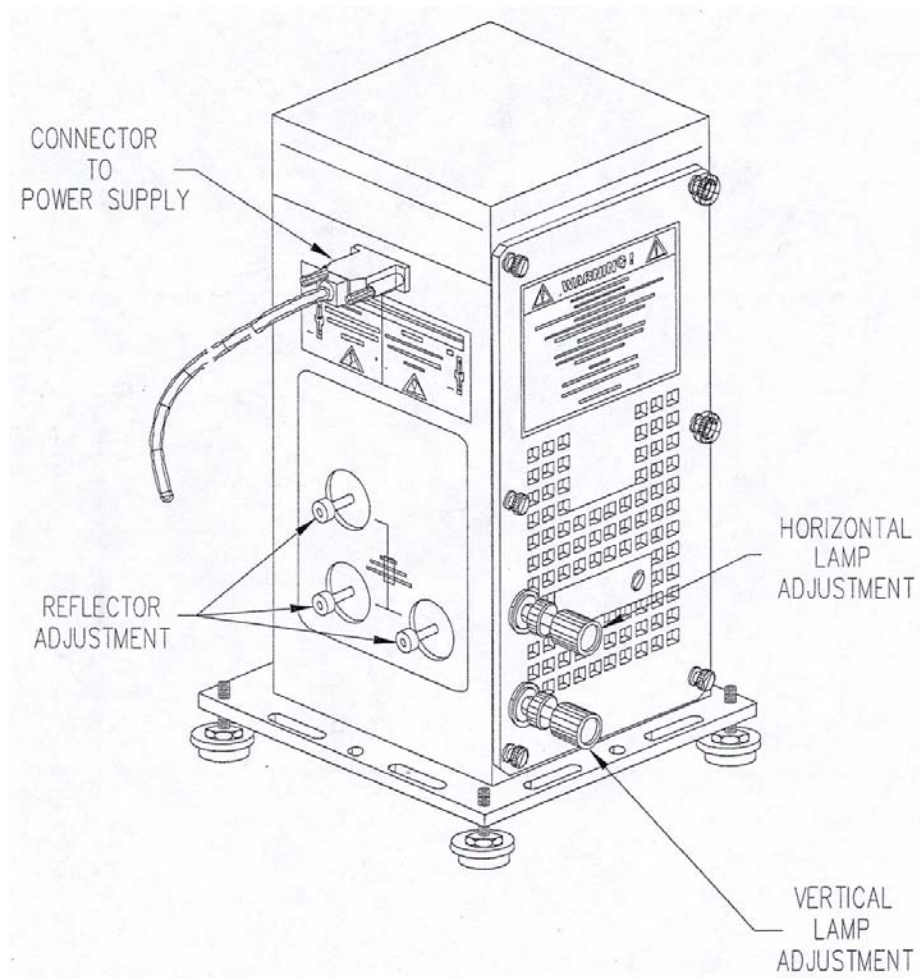


Figure 1

Lamp and Reflector Adjustments

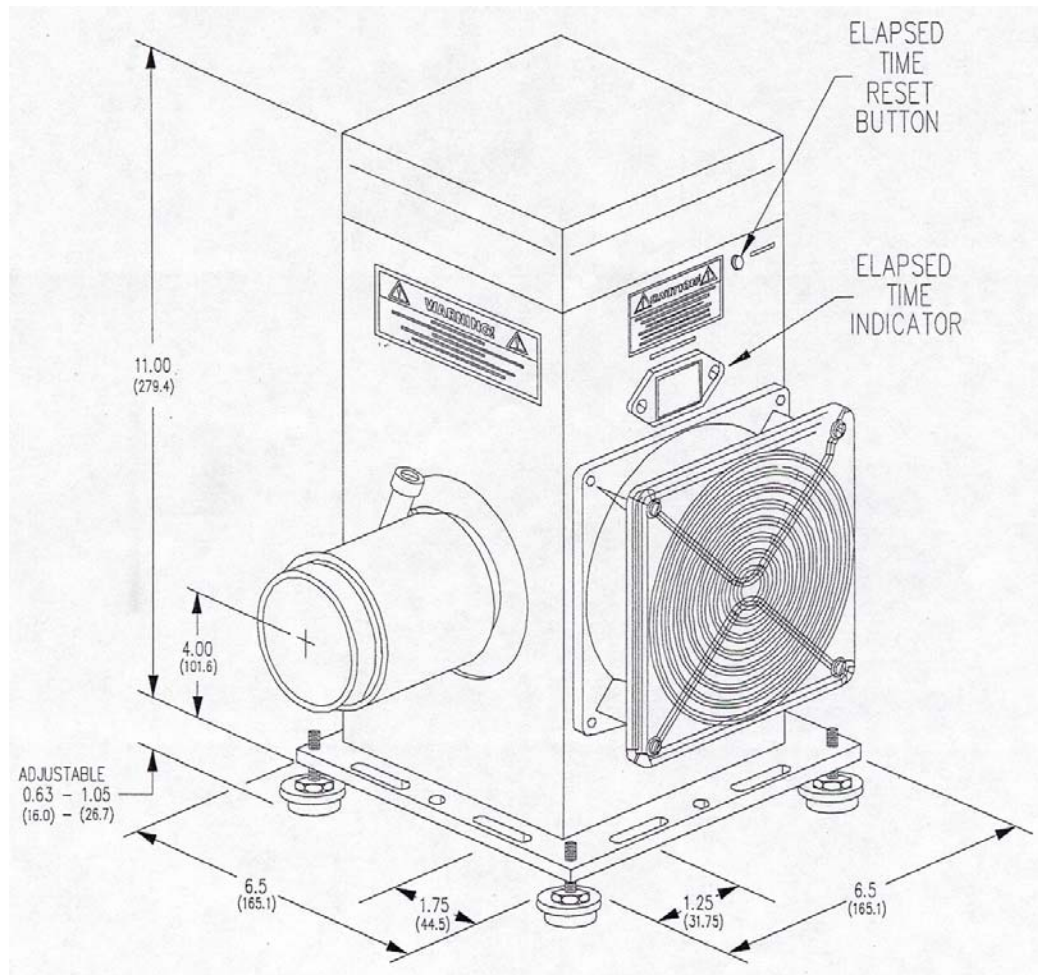


Figure 2
Dimensional Diagram

IV. LAMP INSTALLATION & INITIAL OPERATION

IV.1. INSTALLATION OF THE LAMP

To install the lamp:

- Check that there is no electrical service to the housing and that the lamp power supply is off and discharged.
- Put on safety goggles and gloves.
- Unscrew the six thumb screws on the side of the housing and remove the access door.
- Determine which orientation is correct for the lamp. For xenon and mercury(xenon) lamps, the anode (+ve) is at the top. You can identify the anode by a + stamped on the lamp base. Any lamp identification writing or numbers will be the correct way up when the lamp is held in the correct operating orientation. See Figure 3 for a diagram.
- Connect the sensor block assembly on the top hanging lead to the top adapter, using the hardware supplied with the adapter. Refer to Figure 4 for typical connection information.
- Set the lamp position adjusters at approximately mid range. Move the adjusters while watching the bottom mounting block to find mid range. Place the bottom terminal (with adapter) into the bottom mounting block. Adjust the lamp vertically so the arc gap lies approximately in the center of the mirror, then tighten the knurled set screw on the side of the mounting block.
- After the lamp is in position, clean the envelope with alcohol and lint free tissue. Fingerprints left on the lamp may cause the lamp to explode when lit.
- Replace the side access door and tighten the knurled thumb screws.
- Connect the lamp cable between the lamp housing and the rear of the power supply. Use the black cable to run xenon lamps; Read the Power Supply Instruction Manual before operation.

CAUTION

BE SURE THE LAMP IS POSITIONED WITH THE PROPER ELECTRODE ON TOP AND THE CORRECT CABLE CONNECTING THE POWER SUPPLY AND THE LAMP HOUSING. OPERATION WITH THE WRONG POLARITY WILL IMMEDIATELY DESTROY THE LAMP.

It is important to re-check the lamp polarity. If you are operating the lamp housing for the first time or changing lamp type, **DO IT NOW**. It may save you a lamp. Incorrect polarity will result in rapid destruction of the lamp.

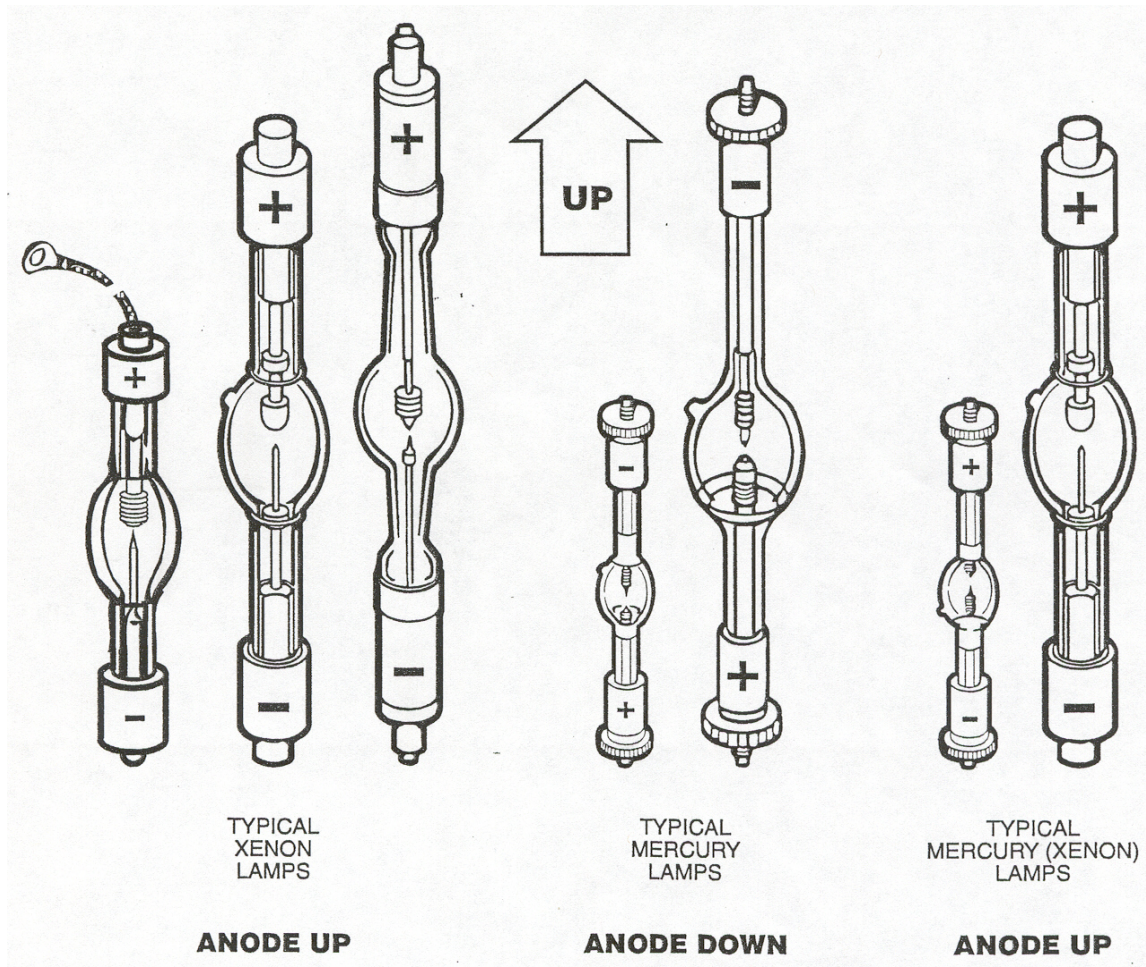


Figure 3

Correct operating position for Arc Lamps

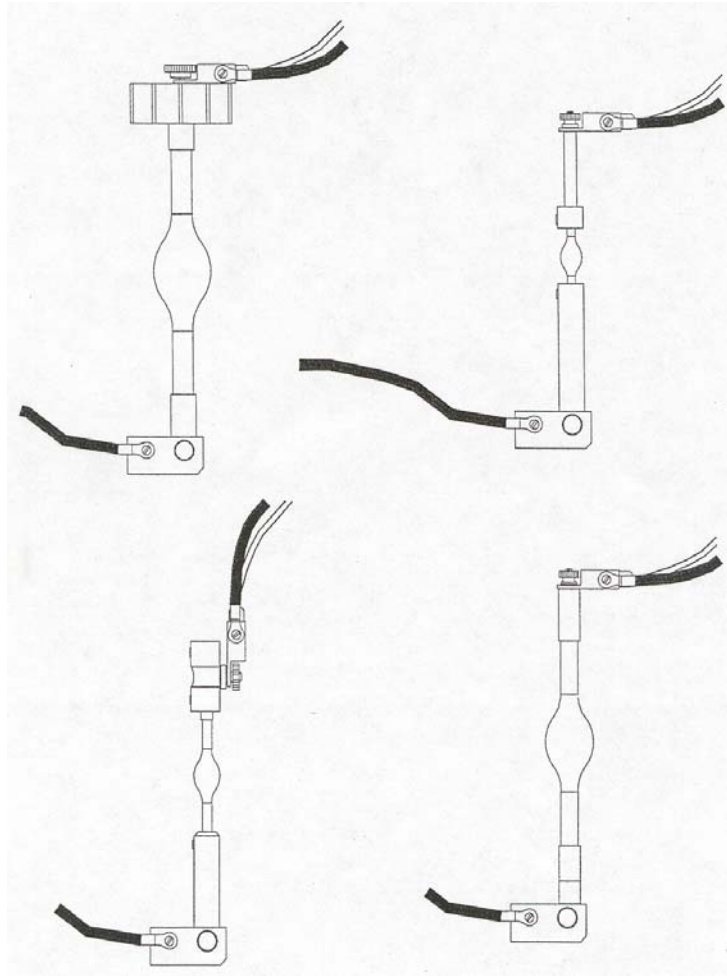


Figure 4

Typical lamp adapter configurations

IV.2. ADJUSTMENT OF LAMP, MIRROR AND LENS

IV.2.1. GENERAL

In these housings, the spherical mirror behind the lamp creates an inverted arc image that can be placed onto or near the lamp arc to increase the output beam intensity. The condensing lens collects radiation from both the arc and the mirror image of the arc.

The mirror is adjusted with respect to the lamp by three knobs (smaller) located directly behind the mirror on the back of the housing. The lamp and mirror in unison are adjusted by a vertical and horizontal adjust knob (larger) in the same area. See Figure 1.

CAUTION

DO NOT LET THE MIRROR IMAGE OF THE ARC FALL ONTO EITHER ELECTRODE. OVERHEATING OF THE LAMP SEALS AND SUBSEQUENT LAMP EXPLOSION MAY RESULT.

IV.2.2. ADJUSTMENTS PRIOR TO OPERATION

1. Before ignition, roughly adjust the lamp position with the horizontal and vertical adjustment knobs. Adjust the lamp so the arc gap lies approximately in the center of the condensing lens. This can be seen through the lens if the focusing lever on the lens is full back.
2. If you have appropriate safety equipment, ignite and warm up the arc lamp per the instructions in the power supply manual.

IV.2.3. ADJUSTMENTS DURING OPERATION

3. By adjusting the condensing lens, focus the output beam so an image of the arc appears on a wall or screen. Do not view UV images without safety glasses.
4. Adjust the lamp position to center the output in the condenser lens aperture. You can place a piece of paper over the condenser output and center the beam on the aperture.
5. Rotate the mirror knobs until a bright spot appears alongside the arc (See Fig. 5 on page). This bright spot is a distorted image of the arc.
6. Focus the image by rotating all three screws in the same direction concurrently (this translates the mirror back or forth). When focused you will see an inverted image of the arc, about the same size as the arc.
7. Place the mirror image over the main (direct) image (as in Fig. 6 on page) or alongside, as desired. You may need to use iterative adjustments to keep a focused mirror image.

CAUTION

YOU CAN SUPERIMPOSE THE TWO IMAGES, BUT AVOID REIMAGING THE ARC HOT SPOTS ONTO THE ELECTRODES. THIS CAN CAUSE OVERHEATING OF THE LAMP.

8. Focus the condensing lens as desired.

If additional range of adjustment of the condensing lens is desired:

1. Remove the condensing lens assembly handle (take care to record the orderly assembly of its parts).
2. By hand, slide the inner lens barrel forward or back until another tapped hole appears in the spiral slot.
3. Reassemble the handle into this new hole.

IV.2.4. LAMP OPERATION AND COOLING

These arc lamps should be operated close to their rated power. Dropping the power below 80% of rated, can lead to unusual lamp performance, eventual instability, and shortened life.

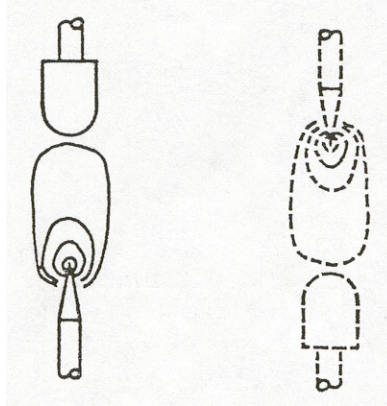


Figure 5

The electrodes as viewed on a screen in front of the condenser lens are inverted (right). The reflector image as a screen (left) is doubly inverted so the anode of a xenon or mercury (xenon) lamp appears on top.



Figure 6

Correctly positioned reflector overlays the inverted arc image on the arc gap. The image is the same size as the arc itself.

IV.3. COOLING

These housings are fan cooled. The fan ensures that the lamp terminals are at an appropriate temperature and the lamp housing skin temperature is safe. The fan speed in these housings is thermally controlled so that the lamp top electrode is kept at a constant temperature as room temperature and lamp cooling requirements vary.

The fan control cycles through four phases:

- lamp warm up phase:** after ignition the fan remains off as the lamp warms up
- fan ramp up phase:** when the lamp reaches a preset temperature the fan comes on at low speed and speeds up as necessary as the lamp continues to warm up until the lamp temperature reaches the control set point
- temperature control phase:** the fan speed is modulated to keep the temperature at the set point
- cool down phase:** when the lamp is turned off, the fan comes on at full speed to cool down the lamp as quickly as possible (as long as the power supply is plugged into the mains voltage supply). The fan will shut off when the lamp has been cooled.

This constant temperature operation minimizes the concern about over or under cooling the lamp unless there are other cooling mechanisms being used - most likely ducting and/or other blowers that are used to remove the ozone by-product that can either augment or restrict the normal cooling air. Before hooking up ducting or other blowers, note the operating voltage and current of the lamp. Check these parameters after adding to your system.

If too much cooling air is provided, the lamp will be overcooled. An overcooled lamp never reaches proper operating conditions and will run with low voltage / high current and often unstable light output. The high current operation will shorten the lamp life. Mercury lamps are very sensitive to overcooling as the mercury does not vaporize.

The evidence of an under-cooled lamp is less obvious. The lamp will warm up quickly, but the voltage and current will be similar to a normal lamp. Since excessive temperature can damage the lamp seals and lead to premature - sometimes dramatic - failure, care has to be taken if the air flow around the housing is changed significantly.

A thermal interlock in these housings will activate and turn off the power supply in a grossly undercooled condition. In the event of a failure in the temperature control circuitry, the fan should run at full speed providing overcooled but safe operation.

V. LAMP HOUSING OPTICS

The collimating lens was designed for efficient collection of light from the arc. In order to get the best performance from your Lamp housing, we first review some aspects of light collection and then describe how to set the lens position.

By moving the focusing lever, you can move the position of the condenser lenses to produce a diverging beam, "collimated beam" or to re-image the arc. The lenses in these housing are designed for collimation rather than imaging. The lens shape and orientation are selected to minimize lens induced distortions (aberration) when the lenses are close to the position that produces a collimated beam (the collimating position). When you use them for imaging, there are two penalties; lens aberrations increase* and light collection is reduced. For imaging, the lens is moved further from the arc, and so gathers less of the light emitted by the arc within its aperture. The lens operates at a higher F/#.

***See Volume III for a comprehensive discussion on aberrations.**

If you need to image the arc close to the Lamp Housing, or equivalently, produce a small image of the arc, then it is more efficient to use the condenser in the collimating position and use a secondary focusing lens to create the image.

To simplify the discussion, we first describe the operation of an ideal lens and then some of the major results of aberrations.

V.1. COLLIMATED BEAMS

The usual concept of a collimated beam is a parallel cylinder of light. If the intensity is the same anywhere across a section of the cylinder, the beam is uniform. Unfortunately, there is no source of a uniform, perfectly collimated beam. Even expensive laser sources have some residual divergence, in the limit governed by the laws of diffraction, and they usually have non-uniform, though sometimes known, intensity distributions.

Arc lamp sources with an ideal condenser lens in the collimating position produce beams that depend on the source size and intensity distribution.

A pinhole source at the focus of an ideal lens produces a beam that is close to the ideal collimated beam. In Figure 7, we show a second pinhole source a distance "d" from the first.

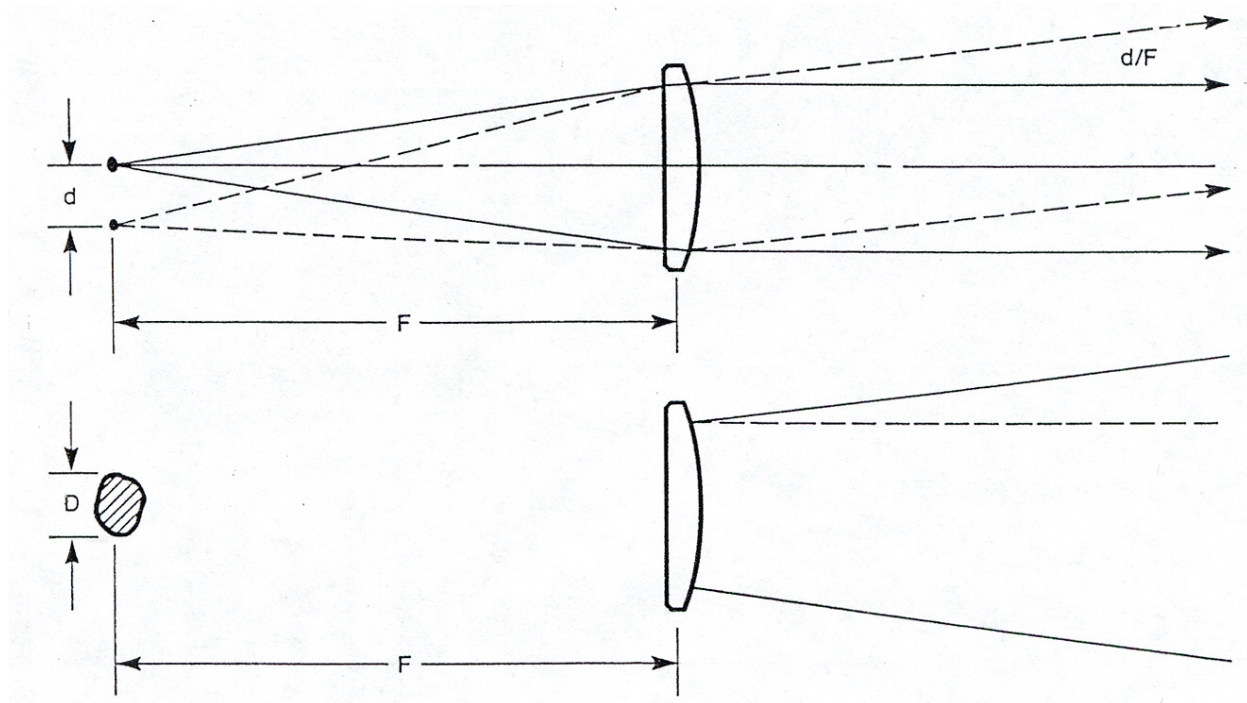


Figure 7

For most arc sources, the divergence in one plane is not the same as that in the orthogonal plane.

The beam from the second pinhole is collimated, but at an angle $\arctan(d/F)$ with the first. Any extended source can be thought of as a whole set of touching pinholes. The beam after the lens is the sum of all the beams from all the pinholes. It will contain rays with angles up to $\arctan(D/F)$ where D is the largest dimension of the source. The beam will have a divergence that depends on the sum of the light from all the points on the source. Obviously, this divergence will depend on the size of the source and the intensity of the various "pinholes" or points on the source.

Most arc sources are non-uniform and are not circular. Therefore, the divergence in one plane is not the same as that in the orthogonal plane. For most design purposes the arc sizes quoted in Volume II and the lens focal length give a good guide to divergence. For low divergence consider the small arc sources and, if necessary, our Spatial Filter Assembly (Volume II).

V.2. UNIFORMITY

Arc sources are not uniform and usually have an intensity peak near one electrode. Intensity contours shown in Volume II indicate arc uniformity. The collimated beam comes from pinhole sources that are not equally intense in addition to being spatially distributed. The result is a smooth, non-uniform beam with some divergence. Figure 8 shows a scan of a collimated beam in the vertical and horizontal planes. The source was a 200W Hg arc lamp.

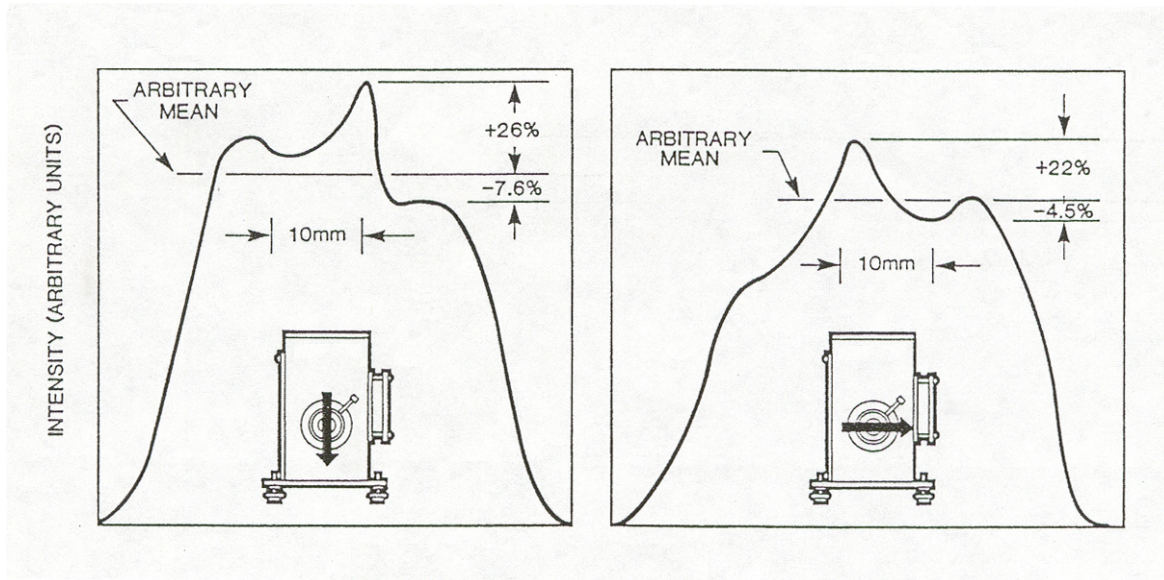


Figure 8

The collimated output of a 200W Hg lamp in the vertical and horizontal planes.

V.3. IMAGING THE ARC

You can reimage the arc by positioning the condenser further from the arc using the focusing lever. Volume III describes imaging and provides the formulae. As the condenser lens is moved out, the image moves in and becomes smaller. As already indicated, the lens collects less light as it is moved away from the arc. Additionally, the convergence angle of the beam goes up as the image becomes smaller. This is not usually important for irradiance of a surface, but can be significant if the image is on the slit of a monochromator, optical fiber, or other optical system with limited acceptance angle. We normally use a secondary focusing lens to maximize the light through a slit or into a fiber optic. Figure 9 shows the higher convergence produced when creating a small arc image.

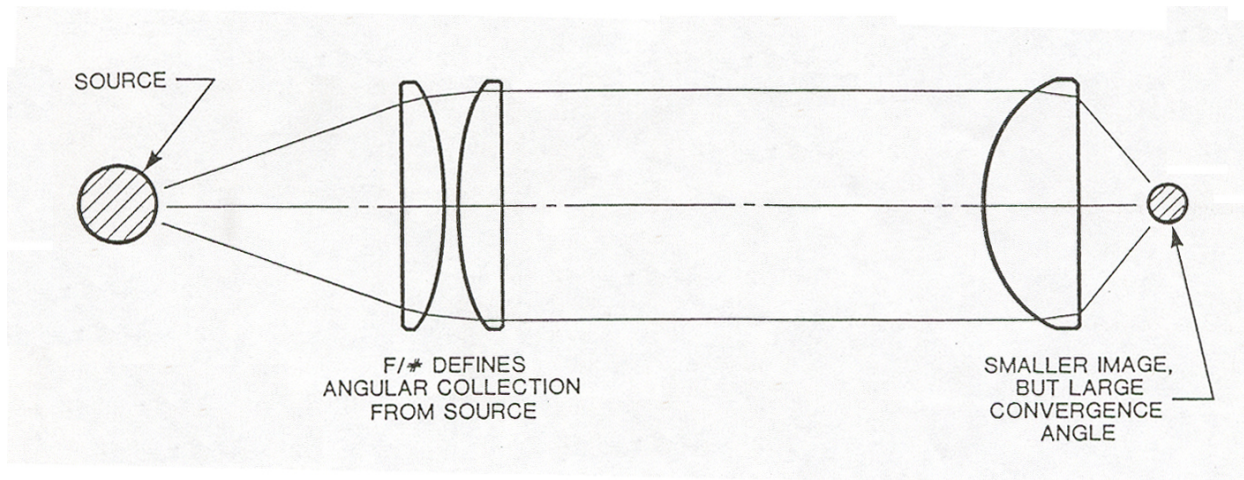


Figure 9

A source focused to a smaller image.

V.4. REAL LENSES

The condenser lenses are intended for efficient light collection. They operate at low F/#s. As a result, the single element F/1 lenses suffers from severe aberrations, particularly spherical aberration.

V.4.1. SPHERICAL ABERRATIONS

This aberration results from the fact that the ideal lens, the aberration free lens, is not spherical in shape. These condenser lenses, like most lenses, have spherical surface shapes for economic manufacturing. In general, spherical aberration is decreased by dividing the refraction (light bending), as equally as possible between as many surfaces as possible. The lens shapes (plano-convex for the fused silica singlets) of our condensers and orientations minimize spherical aberration for the type of condenser and at the collimating position.

Consider the simple plano convex lens collimating light from an ideal point source. With the plano surface towards the point source and the point source at the (paraxial) focus, the marginal rays converge while the paraxial rays are collimated (Fig. 10). This is due to spherical aberration. For the ideal, non-spherical lens shape, the paraxial and marginal rays are all collimated. If the source is located about $1/4 f$ inside the focus, the paraxial rays diverge slightly and the marginal rays are almost collimated. This is often the optimum compromise for a single element collimating lens (and has the added advantage of collecting more light from the source).

The lens adjustment on these lamp housings allows the lens to be moved closer to the source than the paraxial focus. You can empirically find the best position for your system.

Figure 10

The marginal rays and the paraxial rays are collimated when the point source is at the paraxial focus, and the plano surface of the lens faces the point source.

V.4.2. CHROMATIC ABERRATIONS

The term "chromatic aberration" describes the variation of lens focal length with color (Fig. 11). This variation is due to the change in the lens index of refraction (n) with wavelength. As the wavelength goes up, n goes down and the focal length increases.

This causes problems in producing multi-wavelength collimated beams, but is usually a second order effect compared with source and spherical aberration limitations. Chromatic aberration usually becomes significant only when deep UV wavelengths are collimated. The refraction in f (focal length) for a fused silica lens from the visible value of f to $0.91 f$ at 250 nm may require a change of lens for optimum performance. Contact Oriel for details.

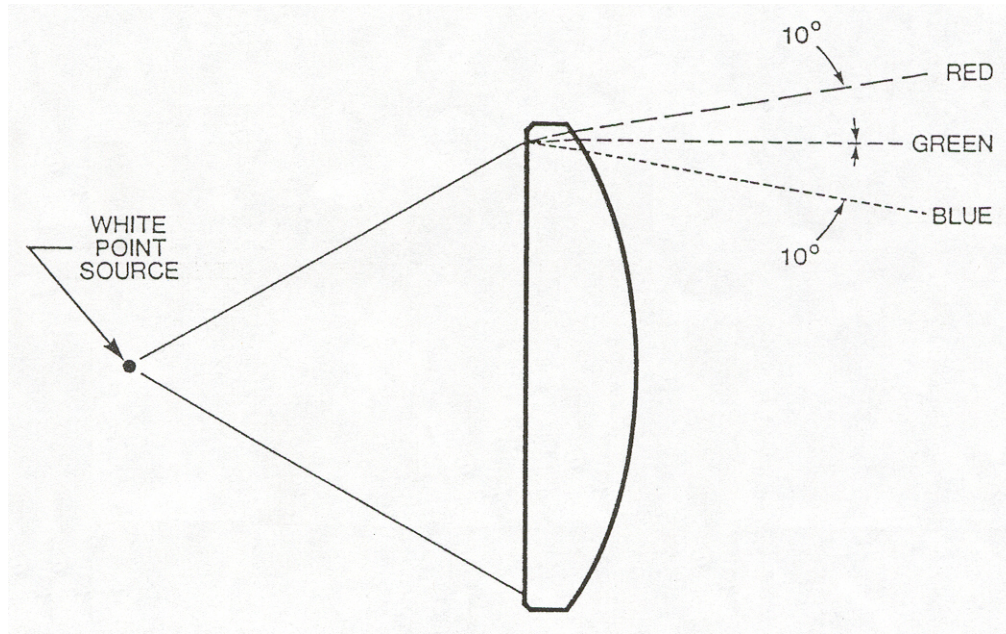


Figure 11

Chromatic Aberration: different wavelengths are focused at different points.

V.5. POSITIONING THE CONDENSER TO GET A COLLIMATED BEAM

You should image the arc on the most distant wall in your laboratory (remembering appropriate safety measures) to get close to the collimation position. You can then move the lens barrel in a small amount for best collimation. If your wall is 2 meters away, a 1.3 mm adjustment is required. For 3 m and 4 m, the corresponding numbers are 0.8 and 0.6 mm.

VI. TROUBLESHOOTING

VI.1. INTRODUCTION

This section deals with procedures to follow if you encounter specific difficulties in operating these Lamp Housings. Additional details for repairing the Lamp Housing circuitry are available from Oriel, but in general, we do not advise attempting to work on the circuitry as this could expose you and your equipment to the high ignition voltage.

VI.2. PROBLEMS

Lamp will not light after several repeated presses of the LAMP START button.

The most common problem experienced when using these Lamp Housings is difficulty in starting the lamp. The problem may be in the Lamp Housing, in the Power Supply, or with the lamp. The following procedure should help you identify the problem area. If you cannot locate the source of the problem, and do not have other Lamp Housings or Power Supplies to interchange as a problem finding technique, we recommend you send the complete system, Power Supply, Lamp Housing and lamp to Acton Research Corporation for diagnosis of the failure mode.

Recommended Procedures

1. Check that the Power Supply is operating -visually inspect the power breaker and make sure the fan is operating.
2. Check that there is pre-ignition voltage available from the Power Supply.

Move the toggle switch on the Power Supply to the voltage position and check that there is an open circuit voltage of at least 60 volts. If not, check the interlock circuit. If there is no pre-ignition voltage then the interlock circuit may not be closed. The principal interlock is the lamp housing door so check that it is closed. There are safety thermostats located in both the lamp housing and the power supply that will break the interlock circuit if a fan fails. It is very unlikely that either thermostat is activated.

If you cannot trace the fault, contact Acton Research for advice. Do not operate the Lamp Housing with a defeated interlock.

3. Check the connection between the Lamp Housing and Power Supply. Make sure the cable is firmly attached and the correct cable for your lamp.

4. Check the lamp and the internal connection to the lamp.

Remove the lamp cable from the Housing, open the Lamp Housing and check that the lamp connections are tight. You may need to remove the lamp and examine it. Assuming no catastrophic damage, then check for cracks in the lamp or lamp stem. Examine the molybdenum strip conductors inside the lamp stem for continuity. Small breaks in these conductors will prevent lamp operation. Examine the electrodes for excessive "burn back" or rounding. If the lamp has a trigger wire, then check that it is properly attached. If it is a xenon or mercury(xenon) lamp, and it does not have a trigger wire, and you would like to attach one, follow this procedure:

- Use a high temperature nichrome wire, AWG 28, with an OD of 0.315 mm.
- Wearing cots or gloves loop the wire around the top of the lamp envelope, (A) in Fig. 12, twisting the end securely.
- Bring the wire down the side of the envelope, (B), nearest to the vacuum "fill/break off" mark. Keep the wire snug against the glass.
- Loop the wire around the snug of the envelope (C).
- Continue this wire, keeping it snug against the glass, down the side of the envelope. Wrap tightly around bottom ferrule and twist the end to ensure it does not slide.

5. Check for ignition.

If all the connections are properly made and the lamps seems fine, then place a piece of paper near the output of the condenser lens and if possible darken the room. Use safety goggles and view the paper while pressing the LAMP START button. What you see will help isolate the problem.

A dim flash, and brief snapping noise indicates that the ignitor is breaking down the lamp, but that the main capacitor in the power supply is not dumping. This indicates that the power supply ignitor drive circuit is working but there is inadequate break down, due either to a problem with the DC output section of the power supply, or a lamp problem.

A brighter flash indicates that the ignitor is breaking down the lamp and the power supply is dumping the output capacitor, but that the power supply is failing to sustain the discharge. The ignitor is operating properly, the lamp or power supply are suspect.

No flash, and no "snap" at all indicates the ignitor or the power supply ignitor circuitry is not working. The ignitor itself, the cabling to the power supply, or the drive circuitry in the Power Supply is at fault.

Depending on the test results, either the Lamp Housing or the Power Supply should be returned to Oriel for repair or replacement.

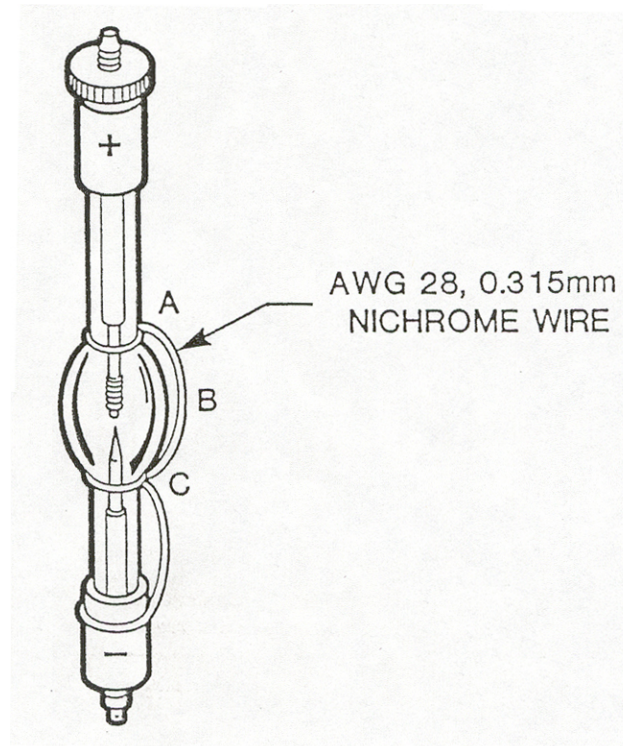


Figure 12

Starter wire installation on Xe lamp.

Lamp Housing Fan does not operate properly.

The Lamp Housing Fan is driven by the lamp's operating temperature and does not start immediately when power is applied. The fan will run until the lamp has cooled off as long as the power supply is plugged in. In most cases, running the Lamp Housing without fan operation can be dangerous. Any malfunction of the fan circuit should be remedied immediately.

Recommended procedure

1. Check that the Lamp Housing cable is connected correctly and that the power supply is connected to an active mains and/or is turned on. If the fan still fails to operate properly, contact Oriel for further assistance.

Elapsed Time Indicator does not operate properly.

The ETI will display the accumulated lamp running hours whenever the Lamp Housing has AC power applied. The meter will count all the time that the lamp is running. Accumulation is indicated by an hourglass symbol at the left of the display.

Recommended procedure

1. Check that the Lamp Housing cable is connected correctly, and the power supply is plugged in. If the ETI still fails to operate properly, contact Oriel for further assistance.

You cannot image the arc properly.

Normally you should be able to image the arc on a surface a few feet or meters away from the Lamp Housing by simply adjusting the condenser lens position. You should also be able to adjust the lamp position to center the beam in the condenser barrel output aperture.

Recommended Procedure:

1. Remove all power from the Lamp Housing and allow the lamp time to cool. Open the Lamp Housing and check the vertical position of the lamp. The arc gap should be positioned in the center of the reflecting mirror. If you have the appropriate adapters for your lamp, you should be able to position it correctly. If not, call Oriel for further information.
2. Check for a broken condenser lens.

You cannot image the arc close to the Lamp Housing.

You should not have a problem imaging the arc some feet or meters away from the Lamp Housing. You may have problems imaging the arc close to the housing because the condenser lens adjustment does not allow the condenser to be located in an appropriate position. As you move the lens out, the image plane moves towards the lens. The closest image plane is determined by how far out you can move the lens.

Recommended Procedure

1. There are three tapped holes in the lens barrel to allow you to change the lens adjustment range. Remove the handle, carefully noting the location of each small part, and reassemble it with the screw in a different position to change the focus range. For reimaging, close to the Lamp Housing, the screw should be in the hole closest to the front (output) of the lens barrel. Note that using a secondary focusing lens, as is done with the XS-433, is often a better approach.

VI.3. INPUT CONNECTOR

Pin assignments for this connector are as follows:

- Pin A1** Power to the bottom of the lamp. Negative voltage for Xe and Hg(Xe) lamps; positive voltage for Hg lamps.
- Pin A2** Power to the top of the lamp. Positive voltage for Xe and Hg(Xe) lamps.; negative voltage for Hg lamps.
- Pin 1** Ground for lamp housing circuitry
- Pin 2** INTERLOCK (+). Connection to +12V required to enable Oriel power supply output. The lamp housing provides this connection through an interlock switch that is closed when the housing door is in place. Oriel power supply operation requires that both INTERLOCK (+) and INTERLOCK (-) be satisfied.
- Pin 3** Power (+12V) for lamp housing circuitry (including fan)
- Pin 4** Ignitor Drive. A ground level (nominal) that activates the lamp ignition circuit in an Oriel series 66900 lamp housing and the “boost” voltage in the power supply when the LAMP START button is pushed. Floating when button not pushed.
- Pin 5** Interlock (-). Connection to GND is required to enable Oriel power supply output. The lamp housing provides this connection through an interlock switch that is closed when the housing door is in place. An overtemperature thermostat is included in the lamp housing circuit to break the interlock and disable the power supply if the housing temperature is too high. Oriel power supply operation requires that both INTERLOCK (+) and INTERLOCK (-) be satisfied.

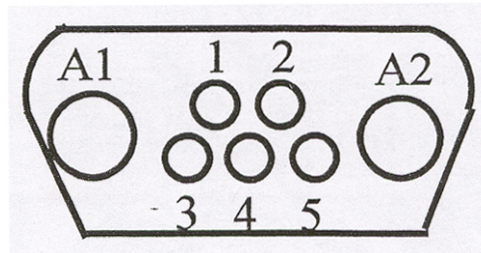


Figure 13

Lamp Housing Connector Pin Assignment

MATING CONNECTOR:

- Body: ITT# DAM-7W2S-A197 (includes pins 1-5)
- Pins: ITT# DM 53744-1 (requires 2 per connector, A1 and A2)
- Backshell: standard 15-pin D-SUB

DECLARATION OF CONFORMITY

Manufacturer's name:	Spectra-Physics
Manufacturer's address:	150 Long Beach Boulevard Stratford, CT 06615 USA
Declares that the product:	
Product Name	A) Arc Lamp Housing
Model Number:	66900, 66901, 66902, 66903, 66904, 66905, 66912
Conforms to the following Product Specifications:	
Safety:	EN 61010-1: 1993 + A2/ IEC 1010-1: 1990 + A1 + A2
EMC:	EN 50081-1: 1992
	EN 55022: 1994 / EN 55011: 1993 Class B
	EN 50082-1: 1992
	IEC 801-2:1991 / IEC 1000-4-2: 1995
	ENV 50140: 1993 / EC 1000-4-3: 1995 / EN 61000-4-3: 1995
	IEC 801-4: 1988 / IEC 1000-4-4: 1995
Complies with the following Directives:	
- the EMC Directive	89/336/EEC
- the Low Voltage Directive	73/23/EEC
And accordingly, carries the CE mark.	
A	
Stratford, CT	December, 1997
	
	George Buzel
	Director of Engineering
	(Signature)
	(Name)
	(Title)

WARRANTY AND RETURNS

Acton Research Corporation warrants that all goods described in this manual (except consumables such as lamps, bulbs, filters, ellipses, etc.) shall be free from defects in material and workmanship. Such defects become apparent within the following period:

1. All products described here, except spare parts: one (1) year or 3000 hours of operation, whichever comes first, after delivery of the goods to the buyer.
2. Spare parts: ninety (90) days after delivery of goods to the buyer.

Acton Research Corporation liability under this warranty is limited to the adjustment, repair and/or replacement of the defective part(s). During the above listed warranty period, Spectra-Physics shall provide all materials to accomplish the repaired adjustment, repair or replacement. Spectra-Physics shall provide the labor required during the above listed warranty period to adjust, repair and/or replace the defective goods at no cost to the buyer ONLY IF the defective goods are returned, freight prepaid, to Acton Research Corporation.

Acton Research Corporation shall be relieved of all obligations and liability under this warranty of:

1. The user operates the device with any accessory, equipment or part not specifically approved or manufactured or specified by Acton Research unless buyer furnishes reasonable evidence that such installations were not the cause of the defect. This provision shall not apply to any accessory, equipment or part which does not affect the safe operation of the device.
2. The goods are not operated or maintained in accordance with Acton Research instructions and specifications.
3. The goods have been repaired, altered or modified by other than authorized Acton Research or Roper Scientific personnel.
4. Buyer does not return the defective goods, freight prepaid, to Acton Research or the Roper Scientific representative within the applicable warranty period.

IT IS EXPRESSLY AGREED THAT THIS WARRANTY SHALL REPLACE ALL WARRANTIES OF FITNESS AND MERCHANTABILITY. BUYER HEREBY WAIVES ALL OTHER WARRANTIES, GUARANTEES, CONDITIONS OR LIABILITIES, EXPRESSED OR IMPLIED, ARISING BY LAW OR OTHERWISE, WHETHER OR NOT OCCASIONED BY ACTON RESEARCH CORPORATION NEGLIGENCE.

This warranty shall not be extended, altered or varied except by a written document signed by both parties. If any portion of this agreement is invalidated, the remainder of the agreement shall remain in full force and effect.

CONSEQUENTIAL DAMAGES-

Acton Research shall not be responsible for consequential damages resulting from misfunctions or malfunctions of the goods described in this manual. The Acton Research total responsibility is limited to repairing or replacing the malfunctioning or malfunctioning goods under the terms and conditions of the above described warranty.

INSURANCE-

Persons receiving goods for demonstrations, demo loan, temporary use or in any manner in which title is not transferred from Acton Research, shall assume full responsibility for any and all damage while in their care, custody and control. If damage occurs, unrelated to the proper and warranted use and performance of the goods, recipient of the goods accepts full responsibility for restoring the goods to their condition upon original delivery, and for assuming all costs and charges.

RETURNS-

Before returning equipment to Acton Research for repair, please call the Service Department at 978-268-0313. Have your serial number available before calling Acton Research. The Customer Service Representative will give you a Return Material Authorization number (RMA). Having an RMA will shorten the time required for repair, because it ensures that your equipment will be properly processed. Write the RMA on the returned equipment's box. Equipment returned without a RMA may be rejected by the Acton Research Receiving Department. Acton Research will notify you of any repairs not covered by the warranty, with the cost of the repair, before starting the work.

Please return equipment in the original (or equivalent) packaging. You will be responsible for damage incurred from inadequate packaging, if the original packaging is not used.

Include the cables, connector caps and antistatic materials sent and/or used with the equipment, so that Spectra-Physics can verify correct operation of these accessories.